The Geometric Dark Sector: Boson Stars as Dark Matter and Vacuum Dynamics as Dark Energy

Jonathan Athanasio Rosa
Independent Researcher

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1 Introduction

The ΛCDM model is a remarkable triumph of modern cosmology: with six parameters it explains CMB anisotropies, large-scale structure (LSS), and the accelerated expansion. Yet, its two dominant components—Cold Dark Matter (CDM) and Dark Energy—remain physically mysterious. Non-gravitational evidence for canonical particle candidates (e.g., WIMPs, QCD axions) is lacking despite decades of direct, indirect, and collider searches [1–4]. Interpreting Λ as quantum vacuum energy leads to the "vacuum catastrophe," a $\sim 10^{120}$ mismatch between naïve QFT estimates and observations.

We advance a geometric interpretation: the dark sector is not an additive substance but an *emergent manifestation* of spacetime geometry and its quantum dynamics. Concretely: (i) Dark Matter is a population of non-baryonic, compact macroscopic objects—*boson stars* (BS)—arising from a primordial scalar field with repulsive self-interaction; (ii) Dark Energy is the residual, dynamical vacuum energy of the quantum gravitational field.

Novelty. While boson stars have been explored as dark-matter candidates before [5–8], this work frames them within a *unified geometric program* that simultaneously addresses dark energy and delivers *specific, falsifiable predictions* across three observational domains—microlensing, gravitational waves, and background expansion—testable by near-future facilities (Roman, LISA, *Euclid*).

2 Observational Predictions I: Microlensing

2.1 Timescale and distribution

For a lens of mass M at distance D_d from the observer and a source at D_s, the (projected) Einstein radius is

$$r_E = \sqrt{\frac{4GM}{c^2}\,\frac{D_d(D_s - D_d)}{D_s}}\,, \qquad t_E = \frac{r_E}{v_\perp} \;\propto\; \sqrt{M}\,, \tag{1}$$

with $v_\perp$ the transverse speed. The *signature* of a BS population is a second component in the t_E histogram (a shoulder/peak) beyond the stellar background.

2.2 Order-of-magnitude targets (Galactic bulge)

Adopting $D_d/D_s \approx 0.5$ and $v_\perp \sim 200\,\mathrm{km\,s^{-1}}$, we obtain:

$$M = 10^{-2}\,M_\odot \Rightarrow t_E \sim 3.5 \text{ days}, \quad M = 10^{-3}\,M_\odot \Rightarrow t_E \sim 1.1 \text{ day},$$
$$M = 10^{-4}\,M_\odot \Rightarrow t_E \sim 8.3 \text{ hours}, \quad M = 10^{-5}\,M_\odot \Rightarrow t_E \sim 2.6 \text{ hours}.$$

2.3 Event statistics and a *strong* prediction

Roman is expected to detect $\mathcal{O}(10^4$–$10^5)$ bulge microlensing events over its survey. If BS constitute a dominant dark-matter fraction ($f_{\rm BS} \sim 1$), the secondary component should be *prominent*, not a subtle feature: a sizable fraction of the total histogram area must appear in the hours–days window above. Even if $f_{\rm BS} < 1$, mixture-model fits to the t_E distribution recover $(f_{\rm BS}, M_{\rm BS})$ with high statistical significance given Roman's cadence and photometric precision.

3 Observational Predictions II: Gravitational Waves

3.1 Frequency band and scaling

For a compact binary, the characteristic GW frequency scales as

$$f \sim \frac{1}{\pi}\sqrt{\frac{GM}{R^3}} = \frac{c^3}{\pi GM}\, C^{3/2}\,, \tag{2}$$

with total mass M, radius R, and compactness $C = GM/(Rc^2)$. For $C \sim 0.1$ and $M \sim 10^{-5}$–$10^{-2}\, M_\odot$, f lies in the mHz–0.1 Hz band—the LISA sweet spot.

3.2 Tidal deformability as a smoking gun

Unlike black holes (which have vanishing tidal deformability, $\Lambda_{\rm tidal} = 0$), boson stars possess *non-zero* $\Lambda_{\rm tidal}$ set by their internal structure. Late-inspiral waveforms will therefore contain measurable tidal-phase corrections and possibly resonant features of internal modes. A confident measurement of $\Lambda_{\rm tidal} > 0$ in this mass and frequency range would be *incontrovertible evidence* for an exotic compact object such as a boson star, and would effectively rule out a primordial black-hole interpretation for the same signal.

4 Observational Predictions III: Dark Energy Dynamics

4.1 Vacuum catastrophe and hypothesis

Naïve QFT estimates give $\rho_{\rm vac} \sim M_{\rm Pl}^4$ whereas observations imply $\rho_\Lambda \sim (10^{-3}\,{\rm eV})^4$. We hypothesize a quantum-gravitational backreaction that cancels the dominant zero-point contributions, leaving a small, positive, *dynamical* residual vacuum energy.

4.2 CPL parametrization and target

We adopt the Chevallier–Polarski–Linder form

$$w(z) = w_0 + w_a\,\frac{z}{1+z}\,, \tag{3}$$

with ΛCDM given by $(w_0, w_a) = (-1, 0)$. **A natural outcome** of a dynamical residual is a deviation at the *percent level*, $|w+1| \sim 10^{-2}$, a key target for Roman/*Euclid*. Joint SNe Ia+BAO+WL analyses can test $w_0 \neq -1$ and/or $w_a \neq 0$ at this precision.

5 Comparison Snapshot and Figures

6 Discussion and Outlook

Theory. Key priorities include: (i) cosmological N-body simulations with BS as discrete components to predict halo profiles and small-scale power; (ii) mapping microphysics (m_ϕ, λ) to the BS mass function; (iii) formation channels in the early universe (e.g., reheating instabilities, fragmentation of scalar condensates).

Domain	ΛCDM/WIMP/BH/NS	Geometric Dark Sector (BS)
Nature of DM	Particle (GeV–TeV)	Boson stars (scalar + GR)
Mass scale	—	10^{-5}–$10^{-2}\,M_\odot$ (illustrative)
Microlensing	Stellar t_E distribution	Second peak/shoulder at 2.6h–3.5d (Roman)
GW band	LIGO/Virgo/KAGRA (10–10^3 Hz)	LISA mHz–0.1 Hz with tidal signatures
Dark Energy	$w = -1$	$w(z) = w_0 + w_a z/(1+z)$; $\lvert w+1\rvert \sim 10^{-2}$

Table 1: Phenomenological contrasts between the standard picture and the Geometric Dark Sector hypothesis.

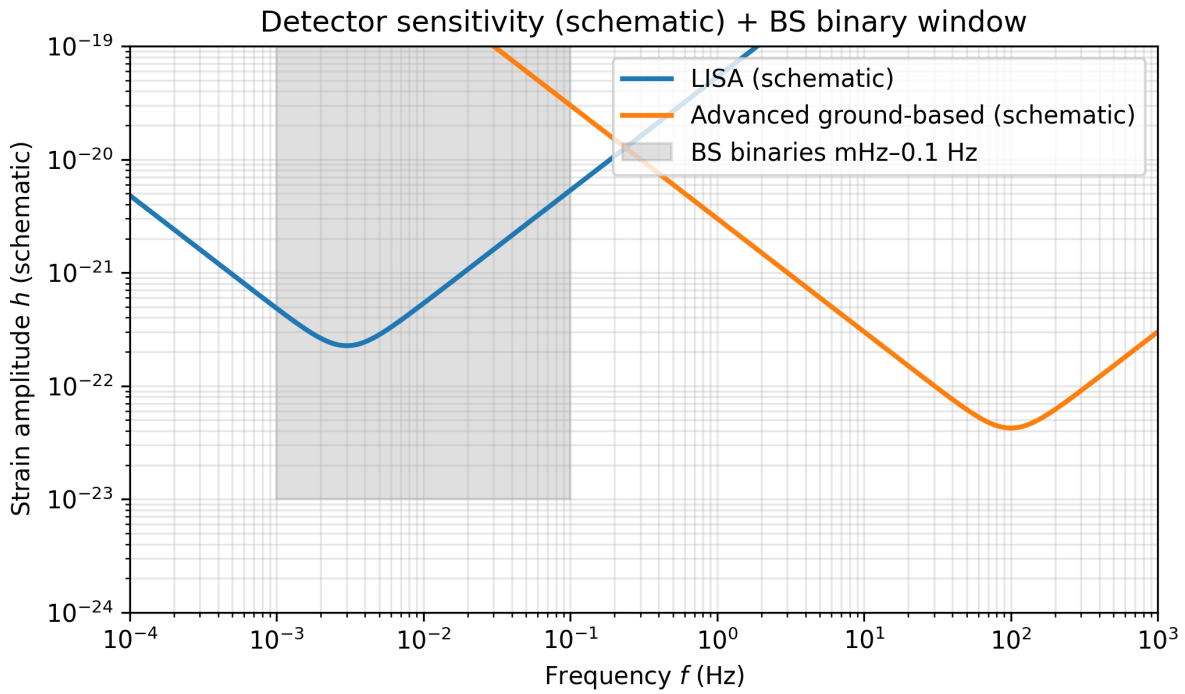

Figure 1: Detector sensitivity (schematic) with the boson–star binary window. The shaded band (mHz–0.1 Hz) falls within LISA's sweet spot. Curves are illustrative (not official sensitivity).

Observations. Degeneracies must be addressed: short t_E events from BS vs. free-floating planets; BS mergers vs. light primordial black holes in GWs. Joint likelihoods across Roman (microlensing), LISA (GWs), and cosmology (SNe+BAO+WL) can break degeneracies.

Synthesis. The Geometric Dark Sector reinterprets 95% of the cosmic energy budget as emergent features of spacetime geometry and quantum dynamics, yielding near-term falsifiable signatures. A single positive detection in any of the three fronts—a robust BS t_E excess, a LISA tidal BS waveform, or $w(z)$ evolution at the percent level—would shift the interpretative landscape.

Falsifiability criteria. Our proposal is decisively testable. It would be *falsified* if: (i) Roman's bulge survey—at projected event statistics—shows no anomalous shoulder/peak in the t_E histogram within the hours–days band compatible with a dominant non-baryonic compact population; (ii) LISA detects no binaries in the mHz–0.1 Hz window *or* all detected systems are consistent with $\Lambda_{\rm tidal} = 0$ (black holes) once tidal-phase terms are constrained; (iii) Roman/*Euclid* confirm $w(z) = -1$ with $|w+1| < 10^{-2}$ across the accessible redshift range. Stating these failure modes emphasizes that the Geometric Dark Sector is a scientific, falsifiable hypothesis.

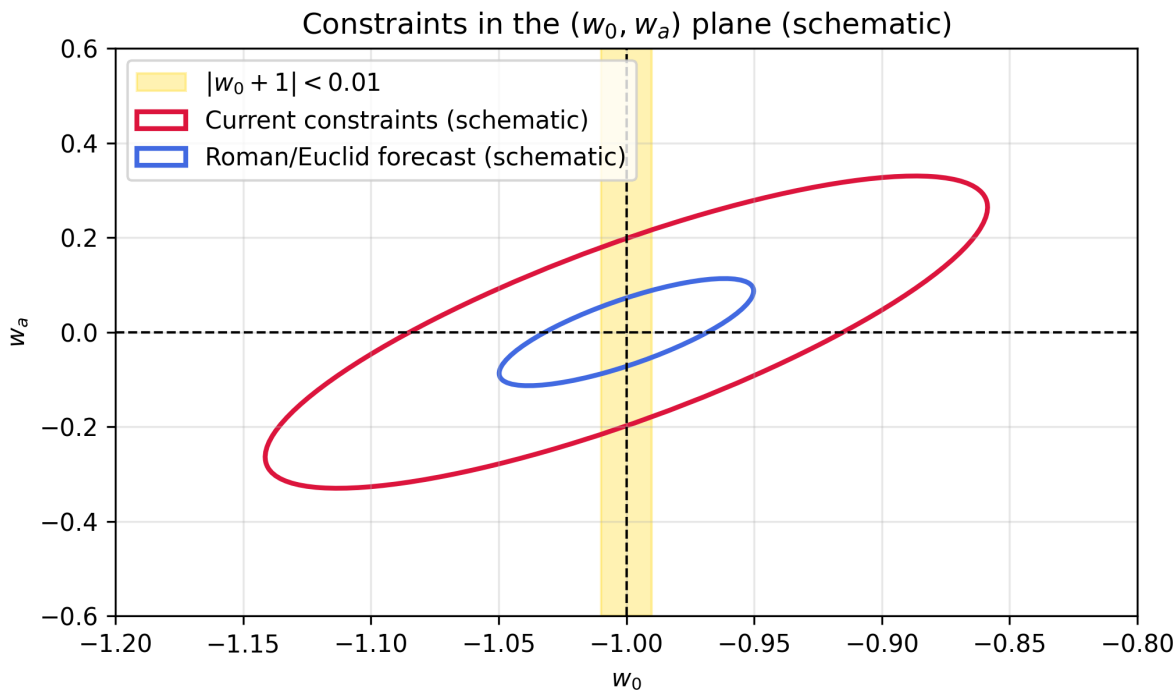

Figure 2: Constraints in the (w_0, w_a) plane (schematic). The yellow band marks $|w_0+1| < 0.01$; detecting $|w_0 + 1| \gtrsim 0.01$ would indicate dynamical dark energy. Ellipses mimic current vs. Roman/*Euclid* forecasts (not official contours).

A Toy Model: Microlensing Event Histogram

To illustrate the expected signal, consider a simplified model in which the differential event rate per logarithmic timescale is given by

$$\frac{dN}{d\ln t_E} \propto \sum_i f_i \exp\left[-\frac{(\ln t_E - \ln \bar{t}_{E,i})^2}{2\sigma^2}\right], \tag{4}$$

where each population i contributes a log-normal distribution centered at mean duration $\bar{t}_{E,i}$ with width σ (here $\sigma \simeq 0.3$ is illustrative). The weights f_i encode the relative optical depth of each lens population.

- Stellar lenses (FGK stars, white dwarfs, brown dwarfs) dominate the "background" component with $\bar{t}_E \sim 10$–30 days.
- A boson-star (BS) population with mass $M_{\rm BS} \sim 10^{-4} M_\odot$ produces $\bar{t}_E \sim 8$ hours.

The resulting histogram exhibits a broad stellar peak plus a secondary shoulder/peak at hours–days, providing a *smoking-gun signature* for BS.

B Compactness and GW Frequency Estimate

For a compact binary of total mass M and radius R, the GW frequency at the innermost orbit is approximated by

$$f_{\rm GW} \approx \frac{1}{\pi}\sqrt{\frac{GM}{R^3}}. \tag{5}$$

It is convenient to express this in terms of the compactness parameter $C \equiv GM/(Rc^2)$:

$$f_{\rm GW} \approx \frac{c^3}{\pi GM} C^{3/2}. \tag{6}$$

Numerically,

$$M = 10^{-2}\,M_\odot,\ C = 0.1\ \Rightarrow f_{\rm GW} \sim 0.1\ \text{Hz},$$
$$M = 10^{-4}\,M_\odot,\ C = 0.1\ \Rightarrow f_{\rm GW} \sim 10\ \text{mHz},$$
$$M = 10^{-5}\,M_\odot,\ C = 0.1\ \Rightarrow f_{\rm GW} \sim 3\ \text{mHz}.$$

These values align with the sensitivity band of LISA. By contrast, black-hole binaries of stellar mass merge in the kHz band, detectable by ground-based interferometers (LIGO/Virgo/KAGRA).

C Appendix C: Toy Plot of the t_E Histogram

We illustrate a minimal, qualitative microlensing duration histogram as a mixture of two log-normal components in t_E (measured in hours): a broad stellar background and a boson-star (BS) component. The stellar peak is placed at $\bar{t}_E^\star \approx 15$ days (~ 360 h), while an illustrative BS population at $M_{\rm BS} \sim 10^{-4} M_\odot$ peaks near $\bar{t}_E^{\rm BS} \approx 8$ h. Amplitudes are arbitrary and meant only to visualize the *shoulder/second-peak* signature.

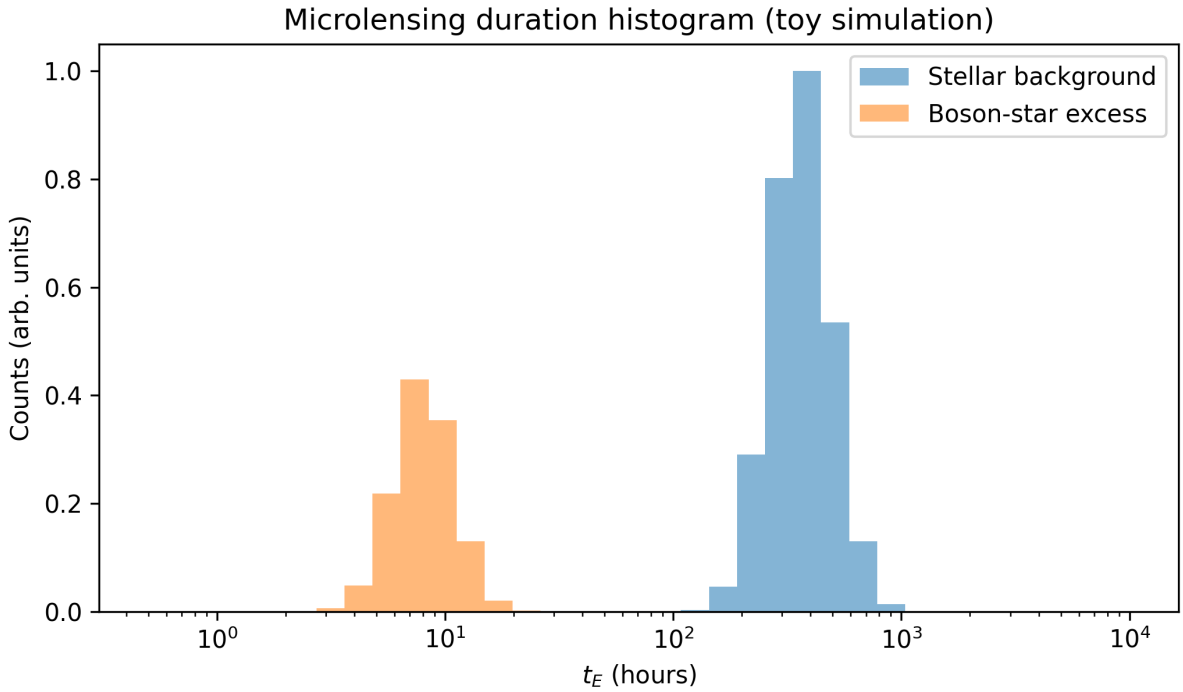

Figure 3: Microlensing duration histogram (toy simulation). The broad stellar component peaks at days, while a boson–star population in the 10^{-5}–$10^{-2}\,M_\odot$ window adds a shoulder/second peak at hours–days, testable by Roman.

D Appendix D: Parameter Space and Tidal Deformability

D.1. Illustrative mass scaling from microphysics

For a self-interacting scalar field with potential

$$V(\phi) = m_\phi^2|\phi|^2 + \tfrac{\lambda}{2}|\phi|^4, \tag{7}$$

the maximum stable boson-star mass follows the Colpi–Shapiro–Wasserman scaling [9]:

$$M_{\rm max} \approx 0.1\sqrt{\lambda}\,\frac{M_{\rm Pl}^3}{m_\phi^2}. \tag{8}$$

Table 2 shows illustrative values for (m_ϕ, λ) and the corresponding $M_{\max}$. We highlight the astrophysically interesting window 10^{-5}–$10^{-2}\,M_\odot$, which yields microlensing durations in the hours–days range (Section 2).

Table 2: Illustrative boson-star mass scales from Eq. (8). Values are approximate, assuming $\lambda \sim 1$.

m_ϕ (eV)	λ	$M_{\max}$ ($M_\odot$)
10^{-22}	1	$\sim 10^{10}$
10^{-18}	1	$\sim 10^{2}$
10^{-11}	1	$\sim 10^{-3}$
10^{-5}	1	$\sim 10^{-15}$

These order-of-magnitude estimates illustrate how ultralight scalars without self-interaction ($\lambda = 0$) lead to either extremely massive, diffuse halos (fuzzy DM regime) or to negligible compact objects, whereas $\lambda > 0$ opens the window to astrophysically compact, testable boson stars.

D.2. Compactness and tidal deformability

The compactness of a boson star is defined as

$$C \equiv \frac{GM}{Rc^2}. \tag{9}$$

Typical solutions with repulsive self-interaction yield $C \sim 0.05$–0.2, smaller than black holes ($C = 0.5$) but comparable to neutron stars. The dimensionless tidal deformability is

$$\Lambda_{\text{tidal}} = \frac{2}{3} k_2 C^{-5}, \tag{10}$$

where k_2 is the Love number ($k_2 \sim 0.01$–0.1 for BS solutions). Thus, even modest deviations in C translate into *large* values of Λ_{tidal} compared to black holes ($\Lambda_{\text{tidal}} = 0$). Detection of a non-zero Λ_{tidal} in the LISA band for sub-solar-mass binaries would therefore be a smoking-gun signature of boson stars rather than primordial black holes.

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